Hackathon IA X RH

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Problem and objective

Context: The company is facing a rising turnover rate and aims to leverage AI to better understand the root causes and retain its top talents.

Objective: An AI solution designed to help HR departments retain their talents.

The chosen themes

AI & Cybersecurity

Secure HR data

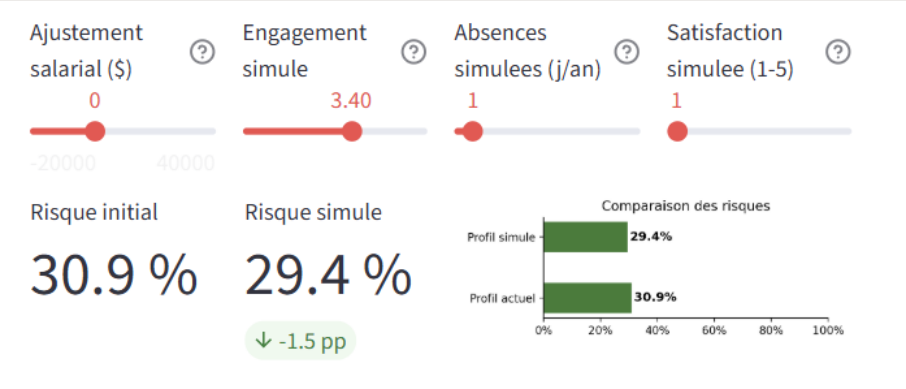
Explainable AI

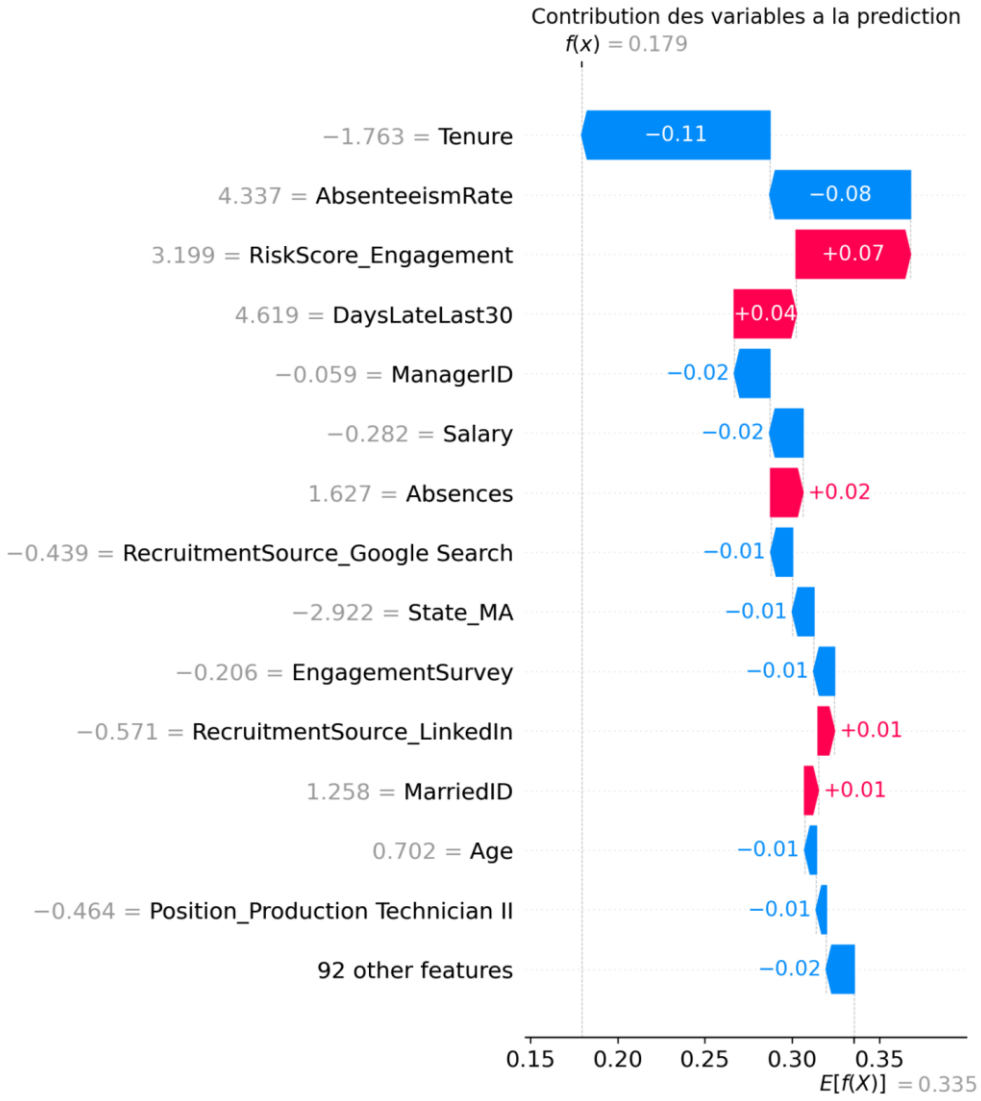
Our solution : The HR Turnover Predictor

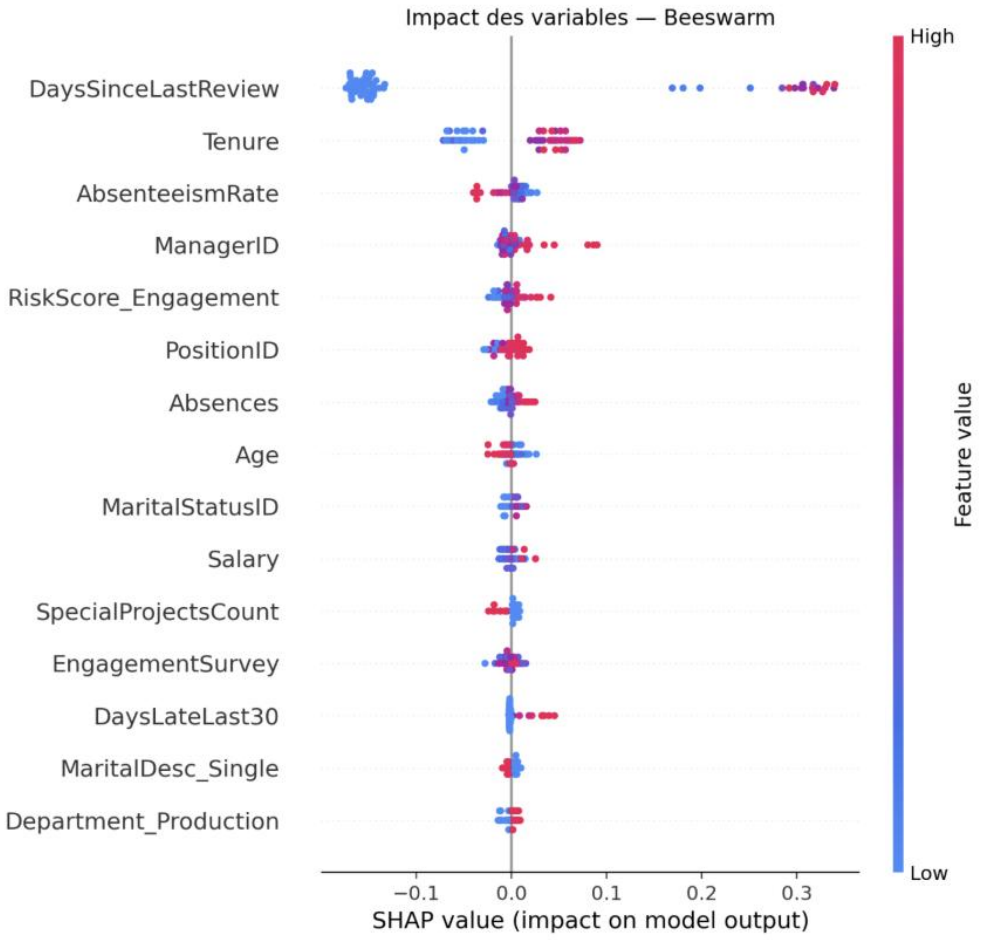
- Individual SHAP Waterfall Plot
- Breaking down exactly why a specific employee is flagged

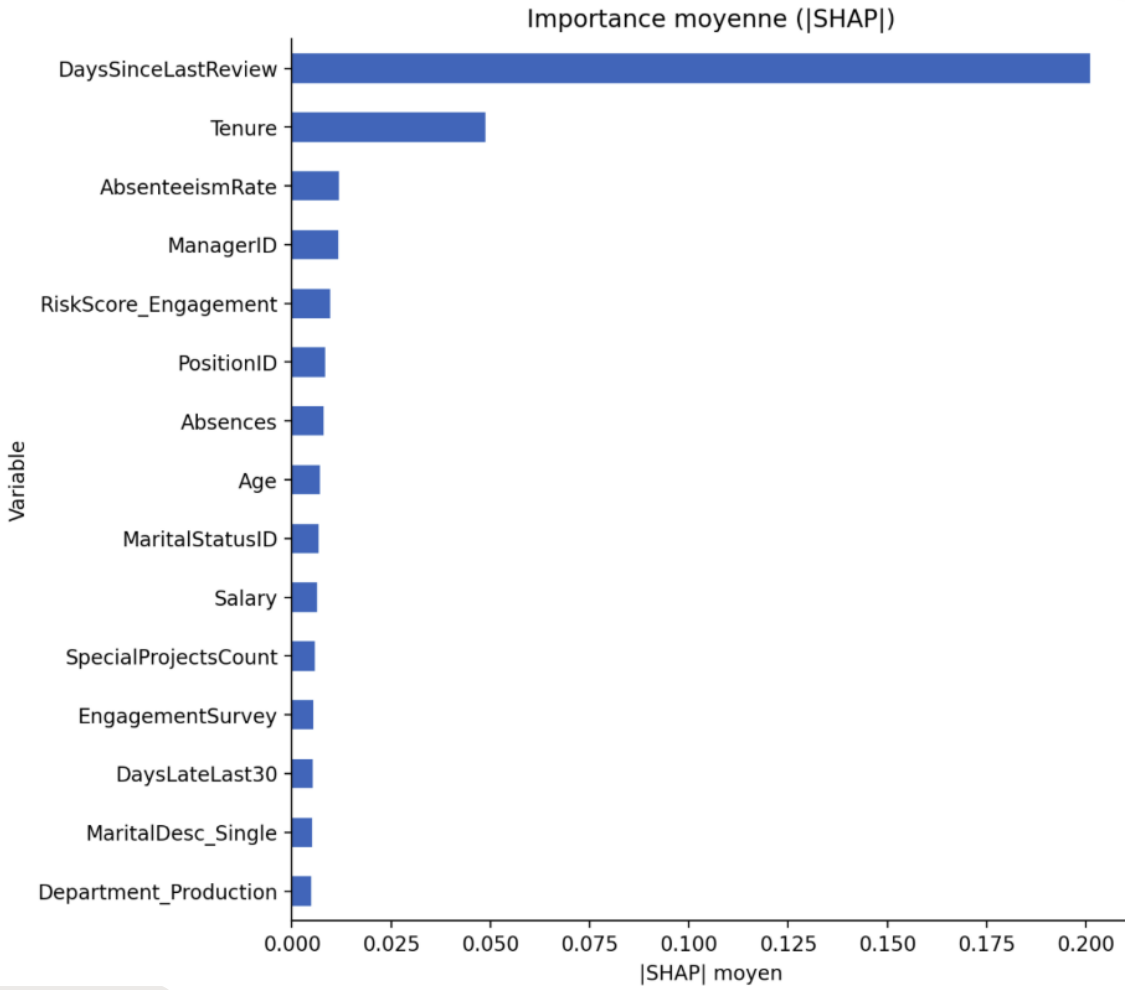
Corrective action simulator : What if ?

- Dynamically testing HR interventions









Explainability AI

Using SHAP to describe how each variable influences each individual decision

AI & Cybersecurity

- **Data Pseudonymization:** Names, identifiers, dates of birth, and zip codes removed.
- **Leakage Variable Removal:** *TermReason*, *EmploymentStatus*, and *EmpStatusID* excluded (post-event).
- **Right to Explanation — GDPR Art. 22:** Each prediction is accompanied by an individual SHAP explanation.
- **Probability Calibration:** Scores interpretable as true probabilities (sigmoid model).
- **Stratified Train/Test Split:** No test data used during training.
- **Equity Audit:** Bias metrics calculated by gender and department.

Variables excluded

	Variable	Raison	Action
0	Employee_Name	Identifiant personnel (PII)	Supprimee
1	EmpID	Identifiant unique (PII)	Supprimee
2	DOB	Date de naissance — remplacee par Age	Transformee en Age
3	Zip	Code postal trop specifique (PII)	Supprimee
4	ManagerName	Identifiant personnel (PII)	Supprimee
5	TermReason	Variable post-evenement (fuite de donnees)	Supprimee
6	EmploymentStatus	Encode la variable cible (fuite)	Supprimee
7	EmpStatusID	Code de statut d'emploi (fuite)	Supprimee
8	DateofTermination	Date de licenciement (fuite + PII)	Supprimee

- **Predictive**: High-accuracy calibrated models
- **Explainable**: Transparent decision-making with SHAP
- **Fair**: Rigorous bias auditing
- **Secure**: GDPR-compliant data processing.

Conclusion : a strategic partner for HR

Demonstration